

7.24 (a) From Eq. (7.13), $\epsilon(-\omega)/\epsilon_0 = \epsilon^*(\omega^*)/\epsilon_0$, let $\omega = -i\omega$, $\omega > 0$, we then have

$$\frac{\epsilon(i\omega)}{\epsilon_0} = \frac{\epsilon^*(i\omega)}{\epsilon_0} \quad \text{Therefore, } \frac{\epsilon(\omega)}{\epsilon_0} \text{ is real on the positive imaginary axis. Following the argument}$$

leading to Eq. (7.14), we can write $\epsilon(\omega)/\epsilon_0$ as

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 - \frac{G'(\omega)}{\omega^2} + \dots,$$

which becomes $\frac{\epsilon(i\omega)}{\epsilon_0} = 1 + \frac{G'(\omega)}{\omega^2}$ on the positive imaginary axis. To establish the desired result, we

need to show $G'(\omega) > 0$. By definition,

$$G(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\frac{\epsilon(\omega)}{\epsilon_0} - 1 \right] e^{-i\omega\tau} d\omega,$$

$$\text{and } G'(\tau) = -\frac{i}{2\pi} \int_{-\infty}^{+\infty} \omega \left[\frac{\epsilon(\omega)}{\epsilon_0} - 1 \right] e^{-i\omega\tau} d\omega.$$

Let $\tau = 0$, we have

$$G'(0) = -\frac{i}{2\pi} \int_{-\infty}^{+\infty} \left\{ \omega \operatorname{Re} \left[\frac{\epsilon(\omega)}{\epsilon_0} - 1 \right] + i\omega \operatorname{Im} \left[\frac{\epsilon(\omega)}{\epsilon_0} \right] \right\} d\omega$$

Using Eq. (7.13) again for real ω , we have

$$\frac{\epsilon(-\omega)}{\epsilon_0} = \frac{\epsilon^*(\omega)}{\epsilon_0}, \text{ or } \operatorname{Re} \left[\frac{\epsilon(-\omega)}{\epsilon_0} \right] + i \operatorname{Im} \left[\frac{\epsilon(-\omega)}{\epsilon_0} \right] = \operatorname{Re} \left[\frac{\epsilon(\omega)}{\epsilon_0} \right] - i \operatorname{Im} \left[\frac{\epsilon(\omega)}{\epsilon_0} \right],$$

which shows that $\operatorname{Re}[\epsilon(\omega)/\epsilon_0]$ is even while $\operatorname{Im}[\epsilon(\omega)/\epsilon_0]$ is odd. Then,

$$G'(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \omega \operatorname{Im} \left[\frac{\epsilon(\omega)}{\epsilon_0} \right] d\omega$$

This is positive, if $\operatorname{Im}[\epsilon(\omega)]$ is positive for all real frequencies

(b)

(c) The Kramers-Kronig relations for $\epsilon_0/\epsilon(\omega)$ are almost identical to those of $\epsilon(\omega)/\epsilon_0$, by the same analyticity consideration, namely,

$$\operatorname{Re} \left[\frac{\epsilon_0}{\epsilon(\omega)} \right] = 1 + \frac{2}{\pi} \mathcal{P} \int_0^{+\infty} \frac{\omega' \operatorname{Im} [\epsilon_0/\epsilon(\omega')]}{\omega'^2 - \omega^2} d\omega',$$

$$\operatorname{Im} \left[\frac{\epsilon_0}{\epsilon(\omega)} \right] = - \frac{2\omega}{\pi} \mathcal{P} \int_0^{+\infty} \frac{\operatorname{Re} [\epsilon_0/\epsilon(\omega') - 1]}{\omega'^2 - \omega^2} d\omega'.$$

Using Eq. (7.159), $\frac{\epsilon(\omega)}{\epsilon_0} = 1 - \frac{\omega_p^2}{\omega^2}$, for large frequency, we can write $\epsilon_0/\epsilon(\omega)$ as

$$\frac{\epsilon_0}{\epsilon(\omega)} - 1 = \frac{\omega_p^2}{\omega^2 - \omega_p^2}, \quad \text{or} \quad \lim_{\omega \rightarrow \infty} \left[\omega^2 \left(\frac{\epsilon_0}{\epsilon(\omega)} - 1 \right) \right] = \omega_p^2. \quad \text{Then, the sum rule similar to}$$

Eq. (7.122) is

$$\omega_p^2 = - \frac{2}{\pi} \int_0^{+\infty} \omega \operatorname{Im} \left[\frac{\epsilon_0}{\epsilon(\omega)} \right] d\omega.$$

(d) In the one-resonance model,

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega},$$

and $\operatorname{Im} \left[\frac{\epsilon(\omega)}{\epsilon_0} \right] = \gamma\omega \frac{\omega_p^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$. By the sum rule, we need to evaluate the integral

$$\gamma\omega_p^2 \int_0^{+\infty} \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} d\omega = \gamma\omega_p^2 \cdot \frac{\pi}{2\gamma} = \frac{\pi}{2} \omega_p^2, \quad \text{which verifies the sum rule.}$$

Similarly, for the same model, $\operatorname{Im} \left[\frac{\epsilon_0}{\epsilon(\omega)} \right] = -\gamma\omega \frac{\omega_p^2}{(\omega_0^2 - \omega^2 + \omega_p^2)^2 + \gamma^2\omega^2}$.

The integral is the same and

$$- \frac{2}{\pi} \int_0^{+\infty} \omega \operatorname{Im} \left(\frac{\epsilon_0}{\epsilon(\omega)} \right) d\omega = - \frac{2}{\pi} \cdot \gamma\omega_p^2 \cdot \frac{\pi}{2\gamma} = -\omega_p^2$$